

05 — Risk Mitigation

TriadicFrameworks: An Open-Science Substrate for Regime-Aware Modeling, Scanning, and Simulation

Submitted to NASA High Priority Open-Source Science (HPOSS)

1. Overview

The TriadicFrameworks project is designed with modularity, transparency, and open-science principles at its core. These characteristics naturally reduce technical and organizational risk. This section outlines the primary risks associated with the project and the mitigation strategies that ensure timely, high-quality delivery of all HPOSS objectives.

2. Technical Risks and Mitigations

Risk 2.1 — Integration Complexity Across Components

TriadicFrameworks includes multiple interoperable components (dsrsp/0.1, RSM, vST, SLRP). Integration across these layers may introduce complexity.

Mitigation:

- maintain strict modular boundaries,
- publish schemas and interfaces early,
- use incremental integration and continuous testing,
- provide synthetic datasets for validation.

This ensures each component can be developed and validated independently.

Risk 2.2 — Variability in Sensor or Simulation Environments

Different sensing systems and simulation engines may produce heterogeneous data structures or noise characteristics.

Mitigation:

- provide a standardized input envelope for dsrsp/0.1,
- include reference examples for multiple modalities,
- supply synthetic test data and reproducible workflows,
- document clear integration patterns.

This reduces the burden on downstream adopters and ensures broad compatibility.

Risk 2.3 — Algorithmic Ambiguity in Regime Classification

Structural and environmental regimes may exhibit ambiguous or overlapping signatures.

Mitigation:

- use transparent, documented classification logic,
- provide example outputs and validation cases,
- include drift and stability indicators to contextualize uncertainty,
- maintain reproducible notebooks for testing and demonstration.

This supports scientific clarity and reproducibility.

3. Schedule and Execution Risks

Risk 3.1 — Delays in Documentation or Integration

Documentation and integration tasks often require more time than anticipated.

Mitigation:

- begin documentation drafting early in the project,
- maintain parallel workstreams,
- use versioned releases to avoid bottlenecks,
- prioritize minimal, clear, and modular documentation.

This ensures steady progress and avoids end-of-cycle compression.

Risk 3.2 — Overextension of Scope

Open-science projects can grow rapidly as new ideas emerge.

Mitigation:

- adhere to a clearly defined 12-month scope,
- prioritize core deliverables (dsrsp/0.1, RSM, vST, SLRP),
- defer non-essential extensions to future releases,
- maintain a public roadmap to manage expectations.

This keeps the project focused and achievable.

4. Sustainability and Community Risks

Risk 4.1 — Long-Term Maintenance Beyond the Performance Period

Sustaining open-source infrastructure requires ongoing attention.

Mitigation:

- use permissive licensing to encourage community adoption,
- publish all materials with DOIs for long-term accessibility,
- provide clear contribution guidelines,
- maintain public repositories with transparent issue tracking.

This supports long-term sustainability and community stewardship.

Risk 4.2 — Limited Early Community Engagement

New frameworks may take time to gain traction.

Mitigation:

- provide high-quality onboarding materials,
- release example integrations for sensing and simulation systems,
- host a virtual workshop during the final project phase,
- maintain active communication channels (e.g., GitHub Discussions).

This encourages early adoption and lowers barriers for contributors.

5. Summary

The TriadicFrameworks project incorporates risk mitigation strategies at every stage of development. Through modular design, transparent documentation, incremental integration, and open-science best practices, the project minimizes technical, schedule, and sustainability risks. These measures ensure that all deliverables will be completed on time, openly accessible, and aligned with NASA's open-science mission.